

Design and Validation of Low Penalty High-Power (>50W) Optical Amplifier for Satellite Communications using Commercial Coherent Transceiver

Rajiv Boddeda⁽¹⁾, Eric Dutisseuil⁽¹⁾, Dinesh Kumar Umasankar⁽¹⁾, Sylvain Bordaïs⁽²⁾, Marco Casale⁽³⁾, Vincent Pollier⁽³⁾, Regis Grasser⁽³⁾, Khmaies Guesmi⁽⁴⁾, Laurent-Daniel Haret⁽⁵⁾, Sébastien Bigo⁽¹⁾

⁽¹⁾ Nokia Bell Labs, Optical Transmission Dept., Massy, France; *rajiv.boddeda@nokia-bell-labs.com

⁽²⁾ Keopsys, 2 rue Paul Sabatier, Lannion, France

⁽³⁾ Bertin, 155, rue Louis-Armand, Aix-en-Provence, France

⁽⁴⁾ Safran Electronics and Defense, 100 avenue de Paris, 91300 Massy, France

⁽⁵⁾ Safran Data Systems, 5 avenue des Andes, 91940 Les Ulis, France

Abstract A high-power optical amplifier for ground-to-satellite optical links was developed using polarization beam combination to alleviate nonlinear effects. We demonstrate net 100-Gb/s transmission with commercial coherent transceivers and experimentally characterize the resulting penalty from the nonlinear distortions: ©2026 The Author(s)

Introduction

High power optical amplifiers (HPOAs) above 50 W are essential building blocks of future free-space optical systems [1,2]. Digital coherent optics is gaining significant interest to support high-capacity communications [4] however high uplink power is required to overcome atmospheric loss under turbulence [3]. Often a single telecom amplifier is insufficient and hence, coherent beam combining (CBC) has emerged as a possible way to scale output power up. Recently, two amplified beams at 50 W were combined to deliver up to 100 W, causing minimal penalties to a 25 Gbps signal [5], but higher-bit-rate signals were impaired by up to 5 dB owing to Kerr nonlinearities even with low entropy modulation such as quaternary phase shift keying (QPSK), unless nonlinear equalization was implemented [6]. The Kerr effect produces phase distortion, spectral broadening, and nonlinear phase noise, and sets an optimal launch power beyond which the signal to noise ratio does not improve further [7,8]. Moreover, CBC requires active phase-locking of all the beams throughout the lifetime of the system. Thus, the interplay between amplifier scaling, beam combining, and nonlinearities is an essential concern for system-design.

High-power amplifiers utilizing very large mode area (LMA) doped fibers are less affected by non-linearity compared to telecom-grade Er-doped fibers [9]. However, prior work employed multi-stage Raman lasers at 1480 nm as pumps, which are limited by low wall-plug efficiency. In this study, we implement amplifiers based on LMA fibers but powered by 980 nm pump diodes. We achieve reduced non-linearity by separately amplifying the two orthogonal polarizations using two rare-earth-doped amplifiers and then combining them in free-space. We evaluate the distortions caused by this scheme to a signal at net rate of 100 Gbps signal (but line rate >200 Gbps) from a coherent dual-polarisation transceiver without any specific non-linear equalization, nor modification of the digital signal processing, thereby demonstrating the readiness of this HPOA for commercial deployment.

Experimental setup

Figure 1 depicts our experiment setup where two polarisation maintaining amplifiers are used to amplify two orthogonal polarisations and combined using a specifically designed high-power multiplexer (HPMUX). A stream of data at

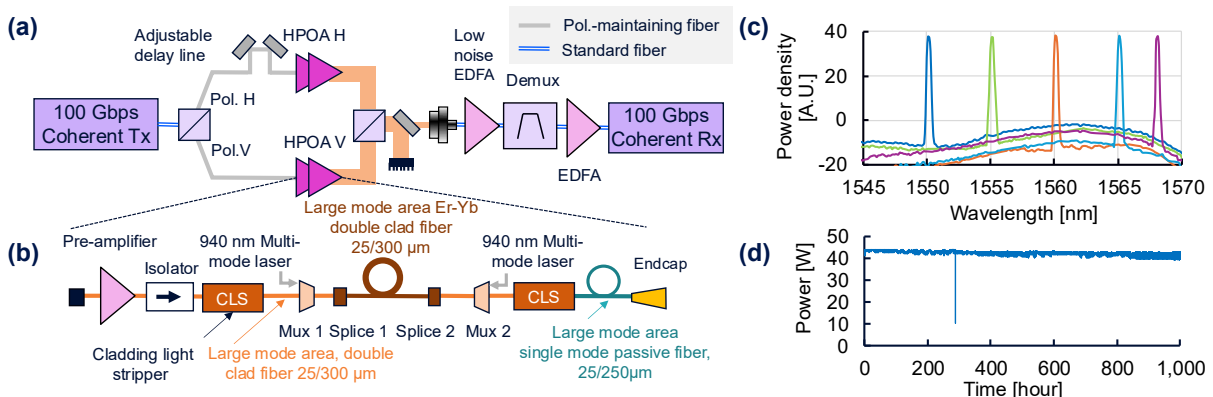


Fig. 1: (a) System architecture. (b) HPOA internal design (c) Optical spectra of HPOA H, fed at different wavelengths between 1550nm and 1570nm (d) Power stability of HPOA H over 6 weeks

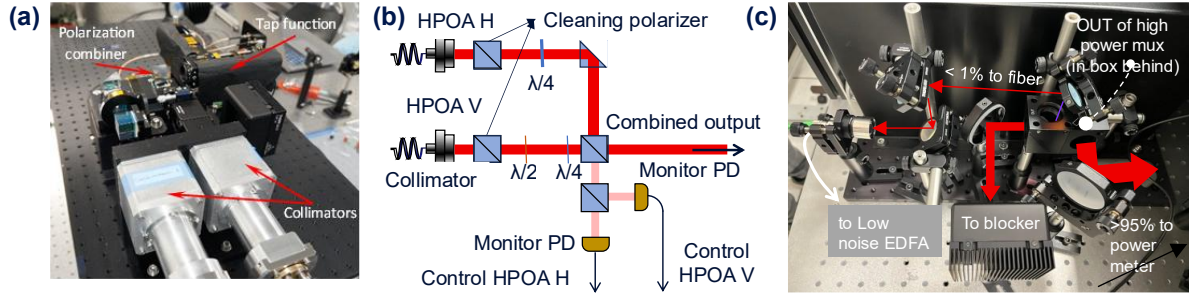


Fig. 2: (a) Breadboard with the HPMUX combination in free space using the two collimators (b) Schematic of the polarization beam combination set-up (c) Attenuation stage at the output of the high-power polarization combiner

~50 Gbaud is modulated with polarization-division multiplexed QPSK, Nyquist shaping, forward error coding with 100% overhead, and detected by commercial-off-the-shelf (COTS) transceivers. To get insight on non-linearity impairments, we substitute these transceivers with their off-line equivalent. The modulated light from the transmitter is decomposed into two arbitrarily oriented, but orthogonal, polarization components (horizontal H and vertical V) in the two arms of a splitter, one arm incorporating an adjustable optical delay line to compensate the path difference between the two arms of the amplification setup. Each component is boosted using a dedicated HPOA before being recombined with the other. The output light travels into a beam splitter where, less than 1% of the power is directed to the receiver through another collimator, 95% is sent to a power meter, while the rest is blocked. This scheme emulates high but typical atmosphere loss. Before coherent demodulation, the receiver chain consists of a low-noise Erbium-doped amplifier (EDFA), a demultiplexer (Demux) to enable WDM operation, and another power-saturating EDFA [10].

High power amplifier design

Each HPOA is embedded in a 3U Rackmount and operated with water cooling. It consists of a pre-amplifier, and a power amplifier, as shown in Fig. 1b. The preamplifier involves two stages and produces 6 W output power. The power amplifier stage is built of Er/Yb co-doped LMA double-clad fiber with core/diameter of 25/300 μm and 0.085 numerical aperture (NA). It is pumped through a power combiner in co- and counter propagation on LMA double-clad passive fiber with core/diameter of 25/300 μm and 0.08 NA. The active fiber is spliced to a matched double-clad passive fiber. Ellipticity, core/clad offset and pump absorption are optimized to suppress higher order modes. The unabsorbed pump power in the double clad fibers is dissipated in cladding-mode strippers. The self-made simple clad passive fiber (25/250 μm , 0.07 NA) incorporates a right-angle endcap with anti-reflection coating at fiber end,

reducing both the power density and the backward power. All components and fibers were designed to maintain polarization. In particular, the end cap is embedded in a FC/PC connector with aligned polarization axis on slow axis in a specific metallic endcap for easy integration. The overall polarization extinction of the HPOAs was measured better than 15 dB while designed gain covered 1550-1568nm bandwidth. The output spectra at 5 different input wavelengths were shown in Fig. 1c confirming the target achievable power of 40 W per HPOA. In fig. 1d, we report the output power evolution of a stand-alone HPOA over 6 full weeks. The stability confirms the absence of any parasitic effects (e.g. no sign of splice failure, photodarkening, 1 μm spontaneous emission nor transverse mode instability).

Polarisation HPMUX and skew correction

We combine the beams from two HPOAs with a high-power polarization multiplexer, which is subset of a larger set-up also involving wavelength multiplexing [11]. Fig. 2a illustrates the collimators, designed with a strong focus on the coupling stability of the HPOAs into subsequent free space. They allow for independent, fine adjustments of beam geometries of the H and V polarization components. They are also designed to reject higher order modes which would otherwise propagate in the system due to the LMA fiber at the HPOA output as it is not fully single mode. These modes would beat with the signal into another parasitic noise and must therefore be filtered out. Importantly, the polarization of these

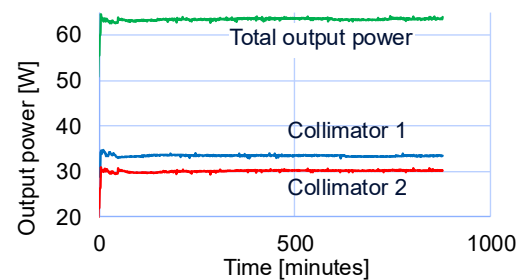


Fig. 3: Overall power evolution of polarization-combining stage with active feedback control

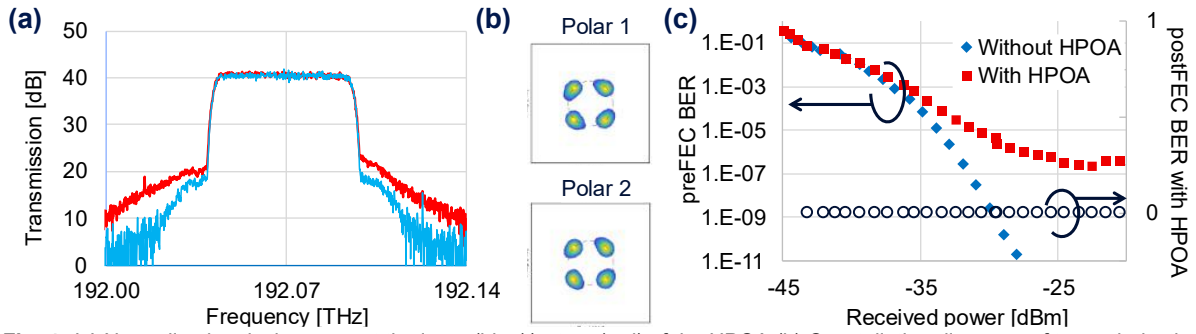


Fig. 4: (a) Normalized optical spectra at the input (blue)/output (red) of the HPOA (b) Constellation diagrams of two polarization tributaries with QPSK format into the amplification testbed (c). Bit-error rate provided by COTS receiver before error correction

modes is orthogonal to the signal polarization. Rather than relying on spatial filtering, we implement mode cleaning with a polarizer prior to multiplexing the two polarization components in the combiner, as schematized in Fig. 2b.

We use the residual leak from the combiner, again split along the two polarization axes, as feedback for the HPOA control. First, the leak provides insight into crosstalk, which we mitigate by adjusting the $\lambda/2$ and $\lambda/4$ waveplates before the combiner. Second, it yields independent readings of the HPOA-V and HPOA-H power levels, which are fed back to the HPOAs for power locking. Fig 3 displays the power evolution after start-up at maximum total power. Stabilisation is obtained withing 1.5% fluctuations 15 min after starting the amplifier. The high-power mux exhibits 0.5 dB (resp. 0.8dB) insertion loss in the paths of collimators 1 and 2, respectively. At full power, we measure a wall-plug efficiency of 7.7%.

Prior to data transmission, we equalize the two polarization arms in the setup. A pulsed laser source is injected at the input while HPOA output power is reduced to 1 W to mitigate the risk of component damage [12]. We measure the differential arrival time of the two pulse replicas using a high-speed oscilloscope and minimize the offset with the delay time. The calibration accuracy is estimated to be 33 ps, with stability better than 10 ps over one week at room temperature. Any residual time skew is compensated by digital signal processing in the receiver.

Coherent Transmission Results

To highlight the spectral distortions due nonlinear effects, we first use coherent transmitters and receivers based on “offline” setup. The setup includes emulating the data generation and algorithms of the commercial coherent receivers at 54 Gbaud, but with off-line processing. Fig. 4a, reports the superimposed optical spectra of the corresponding signals before and after passing through the high-power amplification stage. In the optical spectrum in red, the signature of Kerr nonlinearities under the raised edges can be observed. Fig. 4b reports the obtained constellation

diagrams for both polarisations. They appear distorted with a typical anisotropic signature. The corresponding signal to noise ratio at the receiver is 16.5 dB yielding a bit-error rate (BER) 2.10^{-7} . Lastly, we validate the amplification chain with our COTS transceivers, without any specific tuning. We record the bit-error rate before and after forward error correction and vary the power in the front of the receiver. We report the results in Fig. 4c, while comparing the performance without and with the insertion of the high-power amplification stage. While the axes of the two polarization tributaries of the signal are generally not aligned with the axes of the two HPOAs, digital signal processing recovers them successfully, while tracking and compensating for skew drifts across the two HPOA paths. When the signal hits the receiver in the higher power range, a penalty can be observed, asymptotically transitioning above -35 dBm into a BER floor at 10^{-7} , owing to nonlinear distortions. Importantly, the penalty reduces as power is reduced and becomes negligible below -37 dBm where BER reaches below 10^{-2} . This result is attributed to the large predominance of the amplified spontaneous emission noise from the low noise EDFA at the receiver end over the nonlinearity-induced noise in the typical conditions of uplink satellite communications reproduced here. In the curve of Fig. 4c, the bottom power limit where the receiver is error free after correction is of vital interest because it serves as reference to calculate the minimum launch power which can close the satellite system budget. Advantageously, at this very level, the penalty is found negligible in Fig. 4c. Therefore, every additional dB of power from HPOAs translates into the same increase onto the signal to noise ratio, making HPOA nonlinear impairments harmless.

Conclusions

We designed and validated a high-power optical amplifier based on polarisation combination using commercial off the shelf coherent transceivers in the context of ground to space optical uplink. The system penalty from optical nonlinearities was found negligible on the power budget of the free space coherent optical link.

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